

TE414 Mobile Communication – Tutorial Notes (Lecture 1)

Topic: Evolution of Mobile Communication (1G to 5G & Beyond)

1 Introduction to Mobile Communication

- Ni teknolojia inayoruhusu watu kuwasiliana bila kutumia waya.
 - Inaruhusu **mobility** — mtu anaweza kuzunguka popote lakini bado kuwasiliana.
 - Mifumo ya mwanzo ya mawasiliano ya mobile ilijumuisha:
 - Two-way radio
 - Public land mobile telephone
 - Amateur (HAM) radio
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2 Invention of the First Mobile Phone

- Aliyeanzisha: **Martin Cooper** (Motorola Engineer).
 - Alizaliwa 1928.
 - 1972–73: Alijenga simu ya kwanza ya mkononi (cell phone).
 - Awali ilikusudiwa kwa magari.
 - Ilifanyiwa majaribio 1974.
 - Anajulikana kama “**Baba wa Simu ya Mkononi.**”
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3 Evolution of Mobile Communication

Kizazi cha mawasiliano kimebadilika kutokana na:

- Watu wengi kuhitaji huduma
 - Kiwango kikubwa cha data
 - Mahitaji ya speed na reliability
- Mabadiliko: **1G → 2G → 3G → 4G → 5G → (Future: 6G & 7G)**
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4 First Generation (1G) – Analog Era

Key Points

- Ilianza Japan (1979) – NTT
- Ilienea USA, Europe miaka ya 1980s
- Ilikuwa **analog**
- Service mode: **Voice only**
- Frequency: **800–900 MHz**
- Access: **FDMA**
- Bandwidth: **30 kHz per channel**

Popular 1G Systems

- AMPS (USA)
- NMTS / NMT (Nordic Countries)
- TACS / ETACS (Europe & UK)

Disadvantages

- Poor voice quality
 - Poor battery life
 - Big bulky phones
 - Low security (FM demodulator could decode calls)
 - Limited users & coverage
 - Poor handoff
 - No roaming
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5 Second Generation (2G) – Digital Era

Launched: **1991 in Finland**

Key Features

- Digital system
- Data rate: **Up to 64 kbps**
- Frequency: **900/1800 MHz**
- Access: **TDMA & CDMA**
- Services: SMS, MMS
- Roaming allowed
- Encrypted voice → better security

Disadvantages

- Low data rate
 - Limited mobility
 - Limited smartphone features
 - Unable to support heavy data (video)
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6 2.5G – GPRS (General Packet Radio Service)

- Transition from 2G to 3G
 - Added **packet switching** to GSM
 - Speed: **56–64 kbps**
 - Introduced always-ON internet
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7 2.75G – EDGE (Enhanced Data Rates for GSM Evolution)

- Also called **EGPRS**
 - Speed: **Up to 170 kbps**
 - Considered **pre-3G technology**
 - Requires upgrade in base stations (transceiver units)
 - No core network changes needed
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8 Third Generation (3G) – UMTS Era

Introduced: **Early 2000s**

Key Features

- System: **UMTS**
- Data rate: **384 kbps**
- Switching: **Packet switching**
- Applications:
 - Video calls

- Mobile TV
 - Video conferencing
 - 3D gaming
 - Access: **W-CDMA**
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9 3.5G & 3.75G – HSPA / HSPA+

3.5G – HSPA

- Improves 3G performance
- High Speed Packet Access

3.75G – HSPA+

- Speeds up to **337 Mbps (downlink)**
 - Up to **34 Mbps (uplink)**
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10 3.95G – LTE (Long Term Evolution)

- Based on **OFDMA**
 - Data rate:
 - **Up-link: 170 Mbps**
 - **Down-link: 320 Mbps**
 - Considered **pre-4G**
 - Lead to development of true 4G
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11 Fourth Generation (4G)

Key Features

- Data rate: **100 Mbps – 1 Gbps**
- Technologies: **LTE, WiFi**
- Frequency: **2–8 GHz**
- Characteristics:
 - All-IP network
 - High security

- High battery usage
- Supports high mobility

Meaning of 4G (MAGIC)

- **M** – Mobile multimedia
 - **A** – Anytime anywhere
 - **G** – Global mobility support
 - **I** – Integrated wireless solutions
 - **C** – Customized personal service
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12 Fifth Generation (5G)

Key Features

- Very high speed: ≈ 1 Gbps
 - Very low latency
 - Supports IoT (smart home, smart city, connected cars)
 - Low battery usage
 - High device density
 - Much faster than 4G
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13 Future Generations

6G

- Integrates satellite + 5G
- Global coverage
- Ultra-fast internet
- Smart cities everywhere

7G

- Space roaming
 - Completely wireless world
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Final Summary Table

Generation	Technology Type	Data Rate	Key Features
1G	Analog	30 kHz/channel	Voice only, FDMA
2G	Digital	64 kbps	SMS, MMS, TDMA
2.5G	GPRS	56–64 kbps	Packet switching
2.75G	EDGE	170 kbps	Enhanced GSM
3G	UMTS	384 kbps	Video calls, W-CDMA
3.5G/3.75G	HSPA/HSPA+	Up to 337 Mbps	Turbo 3G
3.95G	LTE	170/320 Mbps	OFDMA
4G	LTE/WiFi	1 Gbps	All-IP, high speed
5G	NR	1 Gbps+	IoT, low latency